



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

First / Second Semester B.Tech. (CBCS) Degree Examinations

Summer Semester – July 2024

ME1003-1 – ELEMENTS OF MECHANICAL ENGINEERING

Duration: 3 Hours

Max. Marks:100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **1kg of steam contains 0.1kg of moisture and 0.9kg of dry steam. Then the dryness fraction is**
A) 10
B) 0.1
C) 90
D) 0.9
2. **----- is not a non-renewable source of energy.**
A) Wind
B) Petrol
C) Flowing water
D) Natural gas
3. **----- valve is used to regulate the flow of steam from the boiler.**
A) Safety valve
B) Feed check valve
C) Steam stop valve
D) Blow-off valve
4. **Absolute pressure =**
A) Gauge pressure – Ambient pressure
B) Gauge pressure / Ambient pressure
C) Gauge pressure + Ambient pressure
D) None of above
5. **Machines which convert mechanical energy into hydraulic energy are called**
A) Turbines
B) Ic engines
C) Compressor
D) Pumps
6. **In De Laval turbine, velocity of steam**
A) Constant
B) Decreases over the blades
C) Increases over the blades
D) None of the above
7. **Suitable turbine for high head application**
A) Pelton wheel
B) Francis turbine
C) Kaplan turbine
D) Parsons turbine
8. **Which of the following is not an internal combustion engine?**
A) 2-stroke petrol engine
B) 4-stroke petrol engine
C) Diesel engine
D) Steam turbine
9. **Number of working strokes per min. for a four stroke cycle engine is _____ the speed of the engine in r.p.m**
A) Equal to
B) One-half
C) Twice
D) Four-times
10. **Worm and worm wheels are used for**
A)) Intersecting axes
B) Inclined axes
C) Right angle axes which don't intersect
D) Right angle axes which intersect
11. **The ratio of tight side tension (T_1) to the slack side tension (T_2) in a belt drive system is given by**
A) $\frac{T_1}{T_2} = e^{\mu\theta}$
B) $\frac{T_1}{T_2} = e^{\frac{\mu}{\theta}}$
C) $\frac{T_2}{T_1} = e^{\mu\theta}$
D) $\frac{T_1}{T_2} = e^{\frac{\theta}{\mu}}$

12. The Freon group of refrigerants are replaced due to
 A) Ozone depletion B) Decrease in COP
 C) Corrosive nature of the refrigerant D) Global warming
13. The _____ property of the refrigerant at both the liquid and vapour states must be very low
 A) Specific heat B) Viscosity
 C) Freezing point D) Boiling point
14. The melting temperature of soft solder is between
 A) 150 °C to 350 °C B) 250 °C to 450 °C
 C) 350 °C to 550 °C D) 0 °C to 50 °C
15. Which of the following process is also known as climb milling?
 A) up milling B) down milling
 C) both up milling and down milling D) none of the mentioned
16. The thickness of the chip in up milling is _____ at the beginning of the cut.
 A) Minimum B) Maximum
 C) Zero D) none of the mentioned
17. What kinds of goods can benefit from programmable automation?
 A) Goods which are produced in job shop production B) Goods which are produced in batch shop production
 C) Goods which are produced in mass production D) Goods which are produced in small scale industries.
18. What kind of automation can produce a wide variety of products?
 A) Fixed Automation B) Programmable Automation
 C) Flexible Automation D) Integrated Automation
19. The work envelope of the Cartesian robot is?
 A) Hemisphere B) Cylindrical.
 C) Rectangular box. D) Circular loop.
20. The work envelope of the cylindrical robot is?
 A) Hemisphere B) Cylindrical
 C) Rectangular box. D) Circular loop.

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Differentiate renewable energy sources with non-renewable sources. b) A steam at 10 bar and dryness fraction 0.98 receives 140 kJ/kg at the same pressure. What is the final state and temperature of the steam? At 10 bar the following values are noted, $T_s=179.9\text{ }^{\circ}\text{C}$ $h_f=762.61\text{ kJ/kg}$ $h_{fg}=2013.6\text{ kJ/kg}$ Specific heat for superheated steam = 2.25 kJ/kgK c) With a neat sketch, explain working of Francis turbine.	04	L*1	1	1
2.	a) Explain with a neat sketch, the working of Babcock and Wilcox boiler. b) Differentiate between reaction and impulse steam turbine.	08 08	L1 L1	1 2	1 1
3.	a) List and explain functions of any four boiler mountings. b) Explain the working of centrifugal pump with neat sketch.	08 08	L1 L1	1 2	1 1
Unit – II					
4.	a) Explain the working of four stroke diesel engine with neat sketch. b) In a crossed belt drive, the difference in tensions between the tight and slack sides of the belt is 1000 N. Find the tensions on the slack and tight sides, if the angle of contact is 160° and the coefficient of friction is 0.3. c) Explain spur and bevel gears.	08 04 04	L1 L3 L1	3 4 4	1 2 1

5. a) Calculate the brake power of a single cylinder four stroke petrol engine which is running at a speed of 200 rpm. The load on the brake drum is 100 kg and spring balance readings 5 kg. The diameter of the brake drum is 1500 mm and the rope diameter is 10 mm.
- b) List the desirable properties of an ideal refrigerant.
- c) Distinguish between soldering & brazing.
- d) What are the advantages and disadvantages of gear drive.

04	L3	3	2
04	L1	3	1
04	L1	4	1
04	L1	4	1

6. a) With a neat sketch, explain the working of vapor compression refrigeration system.
- b) Describe the principle of arc welding with suitable diagram.

08	L1	3	1
08	L1	4	1

Unit – III

7. a) Explain the following lathe operations:

- i) Turning
- ii) Facing

04	L1	5	1
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- b) List and explain various components of CNC. What are the advantages and disadvantages of CNC.

08	L1	5	1
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- c) Define a robot and list the classification of robot based on physical configuration.

04	L1	5	1
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8. a) Explain the following with sketches:

- i) Up milling
- ii) Down milling

06	L1	5	1
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- b) Explain the following types of automation:

- i) Fixed automation
- ii) nFlexible automation

04	L1	5	1
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- c) Explain the application of Robots in material handling and assembly.

06	L1	5	1
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BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
